Data Dictionary

This data dictionary presents the information stored in the CrayCare Cloud Firestore database. Each table identifies the field name, data type, description, and required or default value. It explains in simple terms how the mobile application, backend services, and ESP32 hardware use each field.

The database uses common data types such as string, number, boolean, timestamp, array, map, and null. The table names are written in a readable form, while the field names remain exactly as they appear in the system.

Table: users/{uid}

Field Name	Data Type	Description	Constraints / Default
full_name	string	User's complete display name.	Required
email	string	Email address used for authentication and account recovery.	Required; unique
role	string	Access role. Only one active administrator account is intended; owners operate assigned tank features.	owner | admin
status	string	Controls whether the account may use protected app features.	active | disabled
tank_id	string	Tank owned by the user; normally equal to the owner's UID.	Owner: required; Admin: null
photo_url	string	Main link to the user's profile image. photoUrl is kept only so older app versions can still read it.	Optional
fcmTokens	array<string>	Notification token for each signed-in device.	Default: empty
fcmToken	string	Single-token compatibility field for older app versions.	Legacy; optional
created_at	timestamp	Time the profile document was created.	Server timestamp

Table: users/{uid}/notification_settings/preferences

Field Name	Data Type	Description	Constraints / Default
sound	boolean	Enables notification sound where permitted by the device.	Default: true
vibration	boolean	Enables vibration where permitted by the device.	Default: true
critical	boolean	Enables critical water-quality notifications.	Default: true
warning	boolean	Enables warning-level water-quality notifications.	Default: true
feeding	boolean	Enables feeding reminders and completed/skipped feeding results.	Default: true
sampling	boolean	Enables tank sampling reminders.	Default: true
operational	boolean	Enables actuator, device, and sensor-recovery updates.	Default: true
updated_at	timestamp	Time the preferences were last updated.	Server timestamp

Table: users/{uid}/notif_markers/{key}

Field Name	Data Type	Description	Constraints / Default
markerKey	string	Stable identifier for the reminder or scheduled check.	Required
value	boolean | string | number	Marker state or last processed value.	Required
updated_at	timestamp	Time the marker was last changed.	Server timestamp

Table: hardware_system/currentOwner

Field Name	Data Type	Description	Constraints / Default
uid	string | null	UID of the currently assigned owner.	Active owner UID or null
tank_id	string | null	Tank receiving data and device operations from the assigned hardware.	Must match user's tank_id
assigned_by	string	Administrator who made the assignment.	Admin UID
assigned_at	timestamp	Time the assignment became effective.	Server timestamp

Note: When no owner is assigned, both uid and tank_id are null. If the assigned owner is disabled, both values are cleared together.

Table: tanks/{tankId}

Field Name	Data Type	Description	Constraints / Default
owner_uid	string	UID of the owner responsible for the tank.	Required
current_batch_id	string | null	Identifier of the active production batch.	Optional
stocking_date	timestamp	Date and time the current grow-out cycle was stocked.	Optional
last_sample_date	timestamp	Date of the latest recorded sample.	Optional
sample_count	number	Number of samples recorded for the current cycle.	Default: 0
initial_population	number	Counted population stocked at the beginning of the cycle.	Non-negative integer
initial_total_sample_weight	number	Total weight of the baseline sample.	Optional; grams
initial_total_sample_length	number	Total length of the baseline sample.	Optional; centimeters
is_initialized	boolean	Whether tank and baseline production data have been initialized.	Default: false
created_at	timestamp	Time the tank document was created.	Server timestamp

Table: tanks/{tankId}/sensor_readings/latest

Field Name	Data Type	Description	Constraints / Default
temperature	number	Current water temperature.	degrees Celsius
ph_level	number	Current acidity or alkalinity reading.	pH
dissolved_oxygen	number	Current dissolved oxygen concentration.	mg/L
turbidity	number	Current water turbidity.	NTU
turbidity_air	number	Air/reference reading used by the turbidity integration.	Optional
water_level	number	Current pond water level.	centimeters
feed_level	number	Estimated percentage of feed remaining in the hopper; operational only and excluded from WQA inputs.	0-100 percent
estimated_feed_grams	number	Estimated remaining feed calculated from feed level and calibrated hopper capacity.	Optional; grams
buffered_entries	number	Number of locally buffered readings reported by the ESP32.	Optional
recorded_at	timestamp	Time when the ESP32 collected the sensor reading.	Required

Table: tanks/{tankId}/sensor_readings_history/{YYYY-MM-DD}

Field Name	Data Type	Description	Constraints / Default
date_key	string	Calendar key represented by this daily document.	YYYY-MM-DD
summary_version	number	Schema/version number of the summary computation.	Current: 1
summary_sanitized	boolean	Shows that invalid sensor values have already been removed from the summary.	Required
summary_complete	boolean	Indicates whether the daily summary is ready for optimized reads.	Required
sample_count	number	Number of valid history entries included.	Default: 0
processed_entry_ids	array<string>	IDs of readings already included, preventing the same reading from being counted twice.	Default: empty
temp_min	number	Lowest daily temperature reading.	Optional; degrees Celsius
temp_max	number	Highest daily temperature reading.	Optional; degrees Celsius
temp_avg	number	Average daily temperature reading.	Optional; degrees Celsius
temp_sum	number	Sum used to calculate the daily average for temperature.	Optional
temp_count	number	Number of valid temperature readings included in the daily summary.	Optional
pH_min	number	Lowest daily pH reading.	Optional; pH
pH_max	number	Highest daily pH reading.	Optional; pH

Field Name	Data Type	Description	Constraints / Default
pH_avg	number	Average daily pH reading.	Optional; pH
pH_sum	number	Sum used to calculate the daily average for pH.	Optional
pH_count	number	Number of valid pH readings included in the daily summary.	Optional
DO_min	number	Lowest daily dissolved oxygen reading.	Optional; mg/L
DO_max	number	Highest daily dissolved oxygen reading.	Optional; mg/L
DO_avg	number	Average daily dissolved oxygen reading.	Optional; mg/L
DO_sum	number	Sum used to calculate the daily average for dissolved oxygen.	Optional
DO_count	number	Number of valid dissolved oxygen readings included in the daily summary.	Optional
turbidity_min	number	Lowest daily turbidity reading.	Optional; NTU
turbidity_max	number	Highest daily turbidity reading.	Optional; NTU
turbidity_avg	number	Average daily turbidity reading.	Optional; NTU
turbidity_sum	number	Sum used to calculate the daily average for turbidity.	Optional
turbidity_count	number	Number of valid turbidity readings included in the daily summary.	Optional
waterLevel_min	number	Lowest daily water level reading.	Optional; centimeters
waterLevel_max	number	Highest daily water level reading.	Optional; centimeters
waterLevel_avg	number	Average daily water level reading.	Optional; centimeters
waterLevel_sum	number	Sum used to calculate the daily average for water level.	Optional
waterLevel_count	number	Number of valid water level readings included in the daily summary.	Optional
updated_at	timestamp	Time the daily summary was last maintained.	Server timestamp

Table: tanks/{tankId}/sensor_readings_history/{YYYY-MM-DD}/entries/{docId}

Field Name	Data Type	Description	Constraints / Default
temp_min	number	Lowest temperature reading in the ten-minute period.	degrees Celsius
temp_max	number	Highest temperature reading in the ten-minute period.	degrees Celsius
temp_avg	number	Average temperature reading in the ten-minute period.	degrees Celsius
pH_min	number	Lowest pH reading in the ten-minute period.	pH
pH_max	number	Highest pH reading in the ten-minute period.	pH
pH_avg	number	Average pH reading in the ten-minute period.	pH
DO_min	number	Lowest dissolved oxygen reading in the ten-minute period.	mg/L

Field Name	Data Type	Description	Constraints / Default
DO_max	number	Highest dissolved oxygen reading in the ten-minute period.	mg/L
DO_avg	number	Average dissolved oxygen reading in the ten-minute period.	mg/L
turbidity_min	number	Lowest turbidity reading in the ten-minute period.	NTU
turbidity_max	number	Highest turbidity reading in the ten-minute period.	NTU
turbidity_avg	number	Average turbidity reading in the ten-minute period.	NTU
waterLevel_min	number	Lowest water level reading in the ten-minute period.	centimeters
waterLevel_max	number	Highest water level reading in the ten-minute period.	centimeters
waterLevel_avg	number	Average water level reading in the ten-minute period.	centimeters
feed_level	number	Feed level reading; excluded from WQA model features.	Optional; percent
estimated_feed_grams	number	Estimated feed remaining during the aggregate window.	Optional; grams
recorded_at	timestamp	Original capture time of the completed ten-minute window.	Required

Table: tanks/{tankId}/sensors/{sensorName}

Field Name	Data Type	Description	Constraints / Default
min_value	number	Lower boundary of the configured normal/ideal range.	Required
max_value	number	Upper boundary of the configured normal/ideal range.	Required
critical_value	number	Percentage below which the feed level is critical. This alone does not prevent feeding.	feed_level only
hopper_capacity_grams	number	Calibrated feed mass when the hopper is full; used to estimate remaining grams.	feed_level only
updated_at	timestamp	Time the threshold configuration was last updated.	Server timestamp

Note: The feed level statuses are Normal (>20%), Low (11-20%), Critical (1-10%), and Empty (0%). Feeding is skipped only when feed is empty, unavailable, or estimated grams are below the requested amount.

Table: tanks/{tankId}/actuators/{deviceId}

Field Name	Data Type	Description	Constraints / Default
control_mode	string	Operating mode requested by the owner.	on | off | auto
current_state	string	Actual relay state reported by the ESP32.	on | off

Field Name	Data Type	Description	Constraints / Default
last_changed	number	Time the device state or mode last changed.	milliseconds since January 1, 1970; 0 = never

Table: tanks/{tankId}/actuator_logs/{logId}

Field Name	Data Type	Description	Constraints / Default
actuator_type	string	Device associated with the event.	pump | aerator1 | aerator2
action	string	Human-readable description of the operation.	Required
type	string	Machine-readable operation type.	on | off | auto
logged_at	number | timestamp	Time the operation occurred; older timestamp formats can still be read.	New writes: milliseconds since January 1, 1970

Table: tanks/{tankId}/feeder/status

Field Name	Data Type	Description	Constraints / Default
status	string	Current or most recent feeder state.	idle | checking_feed_level | dispensing | completed | skipped_insufficient | blocked
status_reason	string	Human-readable explanation when feeding is blocked or skipped.	Optional
command_id	string	Manual command correlated with the current outcome.	Optional
dispenseCount	number	Cumulative completed dispensing cycles.	Default: 0
lastSeen	number	Most recent feeder heartbeat/status time.	milliseconds since January 1, 1970
last_dispensed_at	number | timestamp	Time of the last completed dispensing cycle.	Optional
last_dispensed_grams	number	Estimated amount associated with the last completed cycle.	Optional; estimated grams
feed_level	number	Latest hopper level available to the feeder.	0-100 percent
estimated_feed_grams	number	Estimated mass remaining in the hopper.	Optional; grams

Table: tanks/{tankId}/feeder_schedules/{scheduleId}

Field Name	Data Type	Description	Constraints / Default
time	string	Human-readable scheduled time.	H:MM
ampm	string	Meridiem indicator.	AM | PM
timeValue	number	Minutes since midnight, used for ordering and overlap validation.	0-1439

Field Name	Data Type	Description	Constraints / Default
grams	number | null	Estimated feed amount. Null uses the default 20 g.	20-200; multiples of 20
days	array<string> | string mask	Recurring days of the week. Firmware may use a Sunday-first seven-character mask.	At least one active day
enabled	boolean	Whether the schedule is active.	Default: true
isDone	boolean	Older field that becomes true only when the latest feeding was completed. Use last_outcome for the latest result.	For older app versions
created_at	timestamp	Time the schedule was created.	Server timestamp
effective_at_ms	number	Time the schedule was created, edited, or enabled again. Earlier schedule times will not be marked as missed.	UTC time stored as milliseconds
last_outcome	string	Latest confirmed result of the schedule.	completed | skipped_insufficient | blocked | failed
last_occurrence_at	number	Scheduled minute associated with last_outcome.	UTC time stored as milliseconds

Table: tanks/{tankId}/feeder_logs/{logId}

Field Name	Data Type	Description	Constraints / Default
action	string	Human-readable feeding-event description.	Required
type	string	Source/category of the feeder event.	auto | manual | missed | error
status	string	Terminal result reported by the ESP32.	completed | skipped_insufficient | blocked | failed
command_id	string	Originating manual command ID.	Optional
schedule_key	string	Originating schedule document ID.	Scheduled feeds only
schedule_time	string	Scheduled feeding time saved in the history record.	Scheduled feeds only
occurrence_at	number	Original scheduled time, even if the device sends the result later after reconnecting.	UTC time stored as milliseconds
requested_grams	number	Feed amount requested by the command or schedule.	Required for execution
estimated_dispensed_grams	number	Estimated amount based on the feeder motor cycle. It is not an exact weight measured by a scale.	Completed only
amount_basis	string	Shows that the dispensed amount was estimated from the feeder motor cycle.	servo_cycle_estimate
estimated_available_grams	number	Estimated hopper mass available before execution.	Optional
feed_level_before	number	Hopper percentage before dispensing.	Optional; percent
feed_level_after	number	Hopper percentage after dispensing.	Optional; percent
level_change_detected	boolean	Supporting indication of feed movement; not proof of exact grams.	Optional

Field Name	Data Type	Description	Constraints / Default
verification_note	string	Additional interpretation of the before/after feed-level check.	Optional
logged_at	number | timestamp	Original event time. Completed logs alone contribute to Consumption Today.	New writes: milliseconds since January 1, 1970

Table: tanks/{tankId}/feeder_commands/{commandId}

Field Name	Data Type	Description	Constraints / Default
command_type	string	Requested manual action.	feed_now
grams	number | null	Requested amount; null uses the default 20 g.	20-200; multiples of 20
issued_by	string	User who created the request.	Owner UID
issued_at	timestamp	Official time when the feeding request was created.	Server timestamp
expires_at	timestamp	Application deadline; firmware also rejects requests more than 60 seconds old.	Required for new writes

Note: Deletion of a command is not proof of completed feeding. The UI confirms outcomes using a matching command_id in feeder status/log records.

Table: tanks/{tankId}/feeder_notification_receipts/{logId}

Field Name	Data Type	Description	Constraints / Default
uid	string	Recipient owner UID.	Required
push_attempt_claimed_at	timestamp	Time when the backend reserved this feeding result for one notification attempt.	Server timestamp

Note: This record confirms that the backend attempted the push notification. It does not guarantee that the phone received it. The notification is still saved in the app inbox.

Table: tanks/{tankId}/machine_learning_assessments/{assessmentId}

Field Name	Data Type	Description	Constraints / Default
uid	string	Owner UID associated with the assessment.	Required
tank_id	string	Tank evaluated by the assessment.	Required
level	string	Final user-facing assessment after safety and data-availability handling.	Good | Moderate | Poor | Critical | Insufficient
model_level	string	Water-quality class produced by the Random Forest model.	Good | Moderate | Poor | Critical
rule_level	string	Optional rule result used as an additional safety check when recent readings show risk.	Optional
safety_override	boolean	True when the safety floor raises the displayed severity.	Default: false

Field Name	Data Type	Description	Constraints / Default
source	string	Shows whether the result came from the trained model, a backup rule, insufficient readings, or old readings.	Required
model_algorithm	string	Name of the method used to produce the assessment.	RandomForestClassifier | Rule-based | Not applied
model_version	string	Version embedded in the deployed model artifact.	Required when model used
training_data_origin	string	Origin of model-training records, such as synthetic prototype or actual pond history.	Required
label_origin	string	Origin of training labels; documents current prototype limitations.	Required
model_feature_count	number	Number of engineered model inputs.	Required when model used
analysis_window_minutes	number	Historical duration analyzed for the assessment.	Default: 60
data_status	string	Shows whether enough recent sensor history was available for the assessment.	ready | insufficient | stale
source_recorded_at	string | null	Timestamp of the newest actual source reading.	ISO-8601
source_age_seconds	number	Age of the newest source reading at processing time.	Non-negative
confidence	number	Model confidence; omitted or not presented as ML confidence for fallback results.	Model output only
driver	string	Primary parameter driving the assessment.	DO | pH | temp | turbidity | waterLevel | overall
driver_label	string	Human-readable driver name.	Required
driver_value	number	Representative value of the primary driver.	Optional
driver_unit	string	Measurement unit of driver_value.	Optional
driver_min	number	Minimum driver value in the analyzed window.	Optional
driver_max	number	Maximum driver value in the analyzed window.	Optional
problem	string	Short statement of the detected water-quality concern.	Optional
insight	string	Trend-aware explanation for the user.	Optional
action	string	Recommended monitoring or management response.	Optional
concerns	array | map	Primary parameter concerns identified during interpretation.	Optional
secondary_concerns	array | map	Additional lower-priority concerns.	Optional
ts_epoch	number	Numeric processing time used to arrange assessment records in order.	UTC time stored as a number
timestamp	string	Human-readable assessment processing time.	ISO-8601

Note: The assessment needs at least six valid ten-minute records in a row. If the newest reading is more than 20 minutes old, the result becomes Insufficient or Stale. Feed level is not included in the Water Quality Assessment.

Table: tanks/{tankId}/batches/{batchId}

Field Name	Data Type	Description	Constraints / Default
batch_status	string	Current lifecycle state of the batch.	active | harvested | superseded
stocking_date	timestamp	Stocking date; the setup calendar day is Day 0.	Required
harvest_date	timestamp	Date the batch was harvested.	Optional
initial_count	number	Counted population stocked at Day 0.	Non-negative integer
current_count	number	Counted population after recorded mortality and harvest events.	Non-negative integer
harvest_count	number	Cumulative crayfish harvested.	Default: 0
total_mortality	number	Cumulative recorded mortality.	Default: 0
harvest_weight_grams	number	Cumulative recorded harvest weight in grams.	Optional
initial_abw	number	Average body weight from the baseline sample.	Optional; grams
initial_abl	number	Average body length from the baseline sample.	Optional; centimeters
final_abw	number	Final average body weight at harvest.	Optional; grams
final_abl	number	Final average body length at harvest.	Optional; centimeters
days_in_culture	number	Calendar days since stocking, updated at midnight rather than as rolling 24-hour periods.	Default: 0
sample_count	number	Number of non-baseline sampling records.	Default: 0
initial_total_weight	number	Total weight of the baseline sample.	Optional; grams
initial_total_length	number	Total length of the baseline sample.	Optional; centimeters
created_at	timestamp	Time the batch record was created.	Server timestamp

Table: tanks/{tankId}/batches/{batchId}/sampling_records/{recordId}

Field Name	Data Type	Description	Constraints / Default
sampling_date	timestamp	Date the sample was collected.	Required
avg_body_weight	number	Average sample weight: total_weight divided by sample_size.	grams
avg_body_length	number	Average sample length: total_length divided by sample_size.	centimeters
sample_size	number	Number of crayfish physically measured in the sample.	Positive integer
total_weight	number	Combined weight of sampled crayfish.	grams
total_length	number	Combined measured length of sampled crayfish.	centimeters
biomass	number	Estimated tank biomass calculated from current counted population and average body weight.	Estimated grams

Field Name	Data Type	Description	Constraints / Default
live_count	number	Current counted live population used in the biomass estimate.	Counted population
is_baseline	boolean	True for the Day 0 baseline; false for weekly samples.	Default: false
created_at	timestamp	Time the sampling record was created.	Server timestamp

Note: Weekly sampling begins seven calendar days after stocking and continues every seven days from the previous weekly sample. The baseline record is excluded from Week N counting.

Table: tanks/{tankId}/batches/{batchId}/mortality_records/{recordId}

Field Name	Data Type	Description	Constraints / Default
mortality_date	timestamp	Date the dead crayfish were discovered or recorded.	Required
mortality_count	number	Number of mortalities in the event.	Positive integer
created_at	timestamp	Time the record was created.	Server timestamp

Table: tanks/{tankId}/batches/{batchId}/harvest_records/{recordId}

Field Name	Data Type	Description	Constraints / Default
batch_id	string	Parent batch identifier retained in the record.	Required
harvest_date	timestamp	Date the harvest was performed.	Required
harvest_count	number	Number of crayfish harvested.	Positive integer
total_weight_kg	number	Combined harvest weight.	kilograms
abw_grams	number	Average harvested body weight.	grams
created_at	timestamp	Time the record was created.	Server timestamp

Table: notifications/{notifId}

Field Name	Data Type	Description	Constraints / Default
uid	string	UID of the intended recipient.	Required
notif_type	string	Notification category used for grouping and preference checks.	critical | warning | reminder | device_auto | general
title	string	Short notification heading.	Required
body	string	Detailed notification message.	Required
is_read	boolean	Whether the user opened or marked the notification as read.	Default: false
created_at	timestamp	Time the notification record was created.	Server timestamp

Note: Completed and skipped-insufficient feeder logs create inbox records and immediate FCM pushes when the user's Feeding preference is enabled.

Table: sensorIngestion/current

Field Name	Data Type	Description	Constraints / Default
hardwareId	string	Stable identifier of the reporting hardware.	Required
source_tank_id	string	Tank bound to the hardware when the reading was captured.	Required
source_owner_uid	string	Owner bound to the hardware at capture time.	Required
source_assignment_at_ms	number	Owner and tank assignment used to reject data from an old or different owner.	milliseconds since January 1, 1970
captured_at_ms	number	Original capture time when the device clock is trusted.	milliseconds since January 1, 1970
temperature	number	Live temperature value.	Optional per validity
ph_level	number	Live pH value.	Optional per validity
dissolved_oxygen	number	Live dissolved oxygen value.	Optional per validity
turbidity	number	Live water turbidity value.	Optional per validity
turbidity_air	number	Turbidity reference/air value.	Optional
water_level	number	Live pond water-level value.	Optional per validity
feed_level	number	Live hopper percentage.	Optional per validity
estimated_feed_grams	number	Estimated hopper mass.	Optional
buffered_entries	number	Count of device readings waiting in the local buffer.	Default: 0

Table: sensorIngestion/current/history/{docId}

Field Name	Data Type	Description	Constraints / Default
hardwareId	string	Stable identifier of the reporting hardware.	Required
source_tank_id	string	Tank bound at capture time.	Required
source_owner_uid	string	Owner bound at capture time.	Required
source_assignment_at_ms	number	Owner and tank assignment recorded when the sensor data was collected.	milliseconds since January 1, 1970
captured_at_ms	number	Original aggregate completion time.	milliseconds since January 1, 1970
temp_min	number	Lowest temperature reading in the ten-minute period.	degrees Celsius
temp_max	number	Highest temperature reading in the ten-minute period.	degrees Celsius
temp_avg	number	Average temperature reading in the ten-minute period.	degrees Celsius
pH_min	number	Lowest pH reading in the ten-minute period.	pH
pH_max	number	Highest pH reading in the ten-minute period.	pH

Field Name	Data Type	Description	Constraints / Default
pH_avg	number	Average pH reading in the ten-minute period.	pH
DO_min	number	Lowest dissolved oxygen reading in the ten-minute period.	mg/L
DO_max	number	Highest dissolved oxygen reading in the ten-minute period.	mg/L
DO_avg	number	Average dissolved oxygen reading in the ten-minute period.	mg/L
turbidity_min	number	Lowest turbidity reading in the ten-minute period.	NTU
turbidity_max	number	Highest turbidity reading in the ten-minute period.	NTU
turbidity_avg	number	Average turbidity reading in the ten-minute period.	NTU
waterLevel_min	number	Lowest water level reading in the ten-minute period.	centimeters
waterLevel_max	number	Highest water level reading in the ten-minute period.	centimeters
waterLevel_avg	number	Average water level reading in the ten-minute period.	centimeters
routing_status	string	Result used when the data cannot be safely assigned to a tank.	Optional: quarantined
routing_reason	string	Reason the data remains in the temporary receiving area.	Optional